

CY 641: Polymer Synthesis

Course Code	CY 641
Course Name	Polymer Synthesis
L-T-P-C	3-0-0-3
Prerequisites	Consent of the faculty member
Intended for	B.Tech/MS/PhD
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Introduction**
Definition, types of polymers, polymer mechanisms, polymer properties [3 Lectures]
- **STEP GROWTH POLYMERIZATION**
General Theory. Types of step growth polymerization reactions. Gelation [3 Lectures]
- **CARBONYL ADDITION-ELIMINATION REACTIONS**
Polyesters, polyamides, polyimides etc. [6 Lectures]
- **NUCLEOPHILIC SUBSTITUTION REACTIONS**
Epoxy resins. MULTIPLE BOND ADDITION REACTIONS: Polyurethanes, Diels Alder reactions. [3 Lectures]
- **FREE RADICAL POLYMERIZATION**
Chemistry of free radicals. Initiation, propagation, termination mechanisms and kinetics, MWD, Free radical coupling reactions, Living free radical polymerizations [3 Lectures]
- **IONIC POLYMERIZATION**
Cationic and anionic, Addition polymerization by radiation. Group transfer polymerization. [6 Lectures]
- **COPOLYMERS**
Free radical copolymerization, Block copolymers, Telechelic polymers, IPN/SIPN, Ring Opening Polymerization. [3 Lectures]
- **COORDINATION POLYMERIZATION**
Ziegler-Natta catalysts; stereo regular polymers; olefin metathesis; metallocene catalysts. Polyacetylenes. [3 Lectures]
- **Additional Module**
biomaterials, nanomaterials etc.

Text Books

1. George Odian, **Principles of Polymerization**, 4th Edition, , Wiley, 2004.
2. Malcolm P. Stevens, **Polymer Chemistry An Introduction**, 3rd Edition, Oxford University Press, 2011.

Journals

1. Macromolecules
2. Biomacromolecules
3. J. Am. Chem. Soc.
4. MacroLetters
5. NanoLetters
6. Chemistry of Materials